OmicsLite BRSP | Assignment Week 3

Differential Expression Analysis of Lung Adenocarcinoma vs Normal Lung Tissue Transcriptomics Data (GSE10072)

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Background

Cigarette smoking is the leading risk factor for lung cancer. However, the precise molecular mechanism of smoking-induced lung cancer development was still unknown in 2008. Therefore, Landi et al. (2008) conducted gene expression analysis using the Affymetrix Human Genome U133A Array (GPL96) on lung adenocarcinoma (LUAD) and paired noninvolved lung tissue from current, former, and never smokers. The smoking information was validated biochemically. LUAD was chosen because it is the most preeminent subtype of lung cancer occurring in both tobacco smokers and non-smokers. The occurrence in all exposure ranges is ideal for a smoking-induced carcinogenesis study.

NCBI Gene Expression Omnibus (GEO) is one of the public data repositories whose database can be mined to conduct differentially gene expression (DEG) analysis from various transcriptomics research. The microarray data from Landi et al.'s (2008) study can be found with the GEO ID of GSE10072. The data can be analyzed with the embedded web-based tool, GEO2R. However, its capacity is only limited to microarray data and basic plots. Therefore, the R statistical software was used in this study as a better alternative. R can analyze other types of transcriptomics data and generate various data visualizations. R is an open-source language and software environment developed for statistical computing, data analysis, and graphics (R Core Team, 2025).

In this study, the GSE10072 dataset will be analyzed for differential expression between LUAD and normal lung tissue with R, and the gene ontology of the significant genes will be determined.

Methods

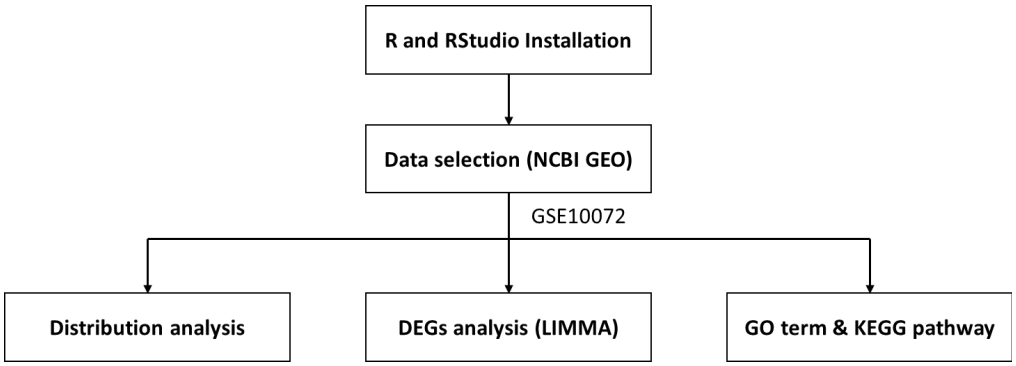

Figure 1. The workflow of this study.

Differential expression analysis was performed using the R statistical software version 4.5.2 (R Core Team, 2025) within RStudio version 20206.01.1+403 (Posit Team, 2025). BiocManager (Morgan & Ramos, 2025) was installed to manage the Bioconductor package.

The microarray dataset with the accession ID GSE10072 was retrieved from the GEO database using the GEOquery package (Davis & Meltzer, 2007), which was later preprocessed by log2 transformation. The GPL96 platform was used for the GSE10072 database, which contains 107 samples, including 58 LUAD and 49 normal lung tissues. Two sample groupings (Adenocarcinoma.of.the.Lung vs Normal.Lung.Tissue) were analyzed for DEGs using the LIMMA (Linear Model for Microarray Data) package (Ritchie et al., 2015). The data was filtered using a fold change of |Log2FC| > 1 and an adjusted p-value of 0.01 calculated with the Benjamini & Hochberg (Benjamini & Hochberg, 1995) false discovery rate (FDR). Gene annotations were mapped according to the hgu133a.db database (Carlson, 2016) using AnnotationDbi (Pagès et al., 2024). Dimensionality reduction was performed with the UMAP algorithm (McInnes et al., 2018) to show data clusters. Data manipulation and visualization were carried out using dplyr (Wickham et al., 2023), ggplot2 (Wickham, 2016), and pheatmap (Kolde, 2019). The top 50 DEGs were submitted to ShinyGo version 0.85 (Ge et al., 2020) to determine their gene ontology.

Result and Discussion

From the GSE10072 dataset, 920 DEGs were identified, with 336 genes (red) upregulated and 584 genes (blue) downregulated in LUAD tissue compared to normal lung tissue (Figure 2). The top 50 DEGs were shown in a heatmap with the annotated genes (Figure 3). All of the genes were downregulated in LUAD tissue except one, secreted phosphoprotein 1 (SPP1). The upregulation of the SPP1 gene is critical in the pathogenesis and progression of diseases, including cancer (Jung et al., 2025). In Yi et al.'s (2022) study, SPP1 was found to be highly expressed in LUAD compared to normal tissue. They discovered that the SPP1 gene facilitates cell migration and invasion by upregulating the downstream COL11A1 gene expression. The upregulation of the COL11A1 gene promotes and accelerates epithelial-mesenchymal transition, which plays an important role in tumor invasion and metastasis.

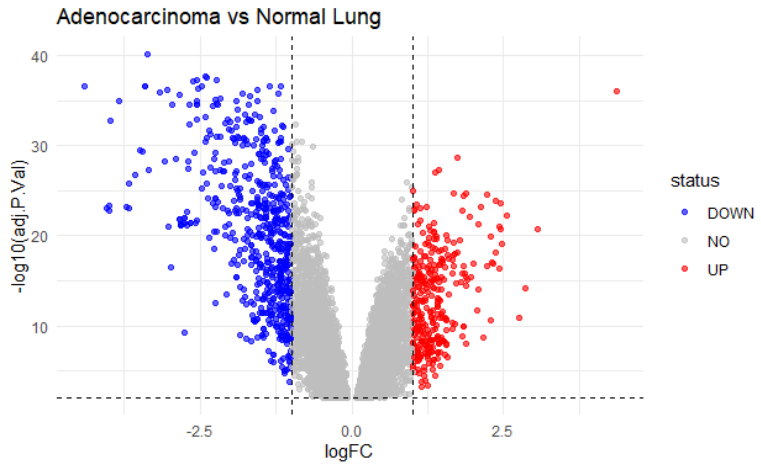

Figure 2. Volcano plot of GSE10072, comparing lung adenocarcinoma and normal lung tissue samples. Blue represents downregulated genes, while red represents upregulated ones.

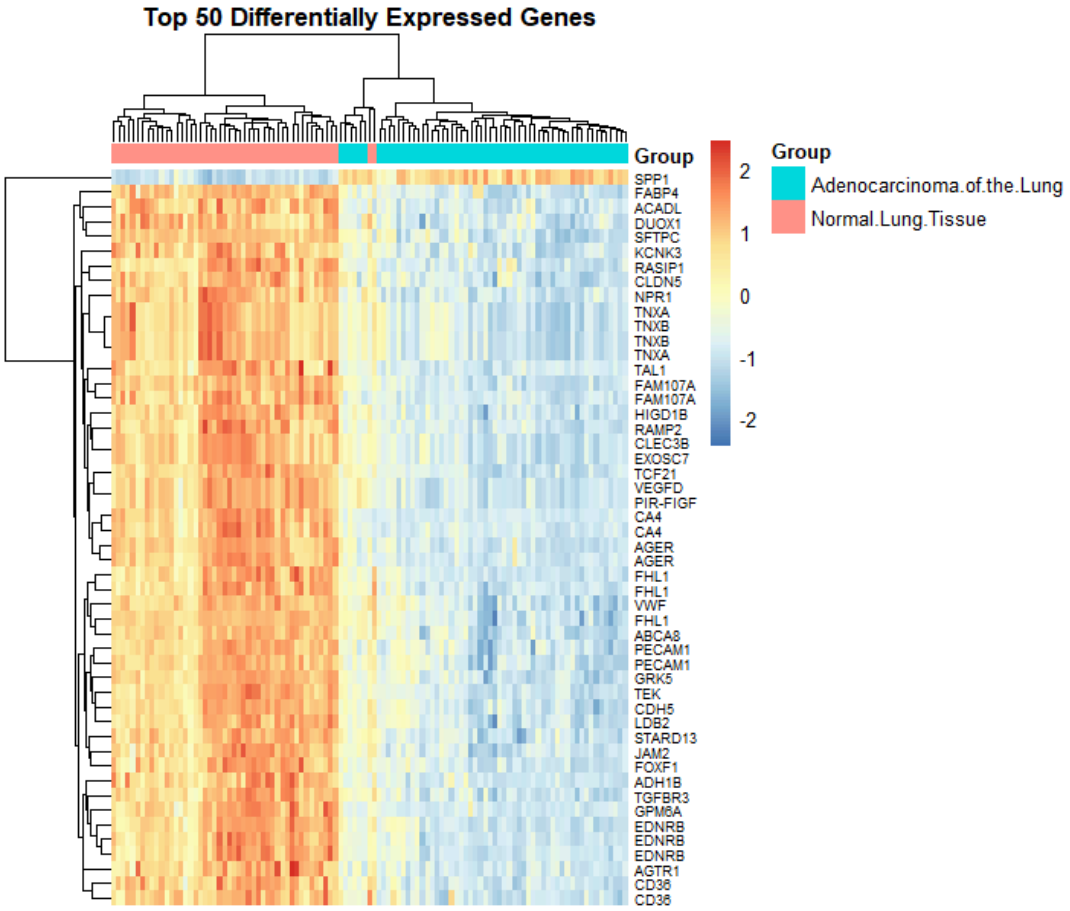

Figure 3. Heatmap of the top 50 DEGs. Blue indicates gene downregulation, while red indicates gene upregulation.

From the GO enrichment result (Figure 4), the term "ECM-receptor interaction" was shown to be the most significant. Followed by leukocyte transendothelial migration, cell adhesion molecules, the PI3K-Akt signaling pathway, and focal adhesion forming the top five significant pathways.

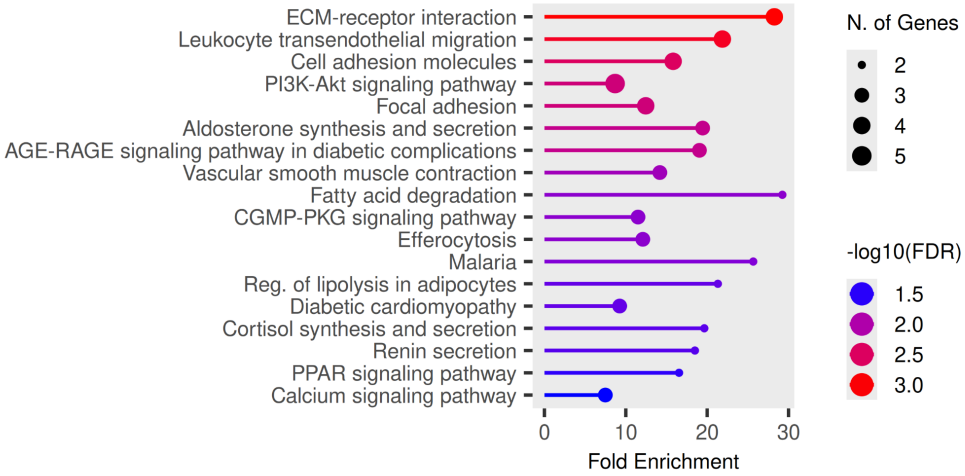

Figure 4. Lollipop chart of the top 50 DEGs sorted by enrichment FDR < 0.05.

The top-ranked five GO terms and KEGG pathways were identified with STRING-db embedded in ShinyGo (Szklarczyk et al., 2022) (Table 1). The term "ECM-receptor interaction" also came out to be the most significantly enriched in the KEGG pathway. Followed by leukocyte transendothelial migration, cell adhesion molecules, the PPAR signaling pathway, and the PI3K-Akt signaling pathway. In the biological process, DEGs were mainly involved in positive regulation of biological process (GO:0048518 and GO:0048522), regulation of biological quality (GO:0065008), vascular process in circulatory system (GO:0003018), and regulation of vascular permeability (GO:0043114). In the cellular component, DEGs were involved in cell periphery (GO:0071944), intrinsic component of plasma membrane (GO:0031226), integral component of plasma membrane (GO:0005887), cell surface (GO:0009986), platelet alpha granule (GO:0031091), and plasma membrane (GO:0005886).

Table 1. List of the top five significantly (p-value < 0.05) annotated GO terms and KEGG pathways by STRING available in ShinyGo 0.85.

FDR	Number of DEGs	GO ID	GO Term
Biological Process (BP)			
1.04e-07	32	GO:0048518	Positive regulation of biological process
1.04e-07	26	GO:0065008	Regulation of biological quality
7.06e-06	9	GO:0003018	Vascular process in circulatory system
9.67e-05	27	GO:0048522	Positive regulation of cellular process
1.5e-04	5	GO:0043114	Regulation of vascular permeability
Cellular Component (CC)			
0.008	25	GO:0071944	Cell periphery
0.0131	13	GO:0031226	Intrinsic component of plasma membrane
0.0185	12	GO:0005887	Integral component of plasma membrane
0.0185	9	GO:0009986	Cell surface
0.0185	4	GO:0031091	Platelet alpha granule
0.0234	22	GO:0005886	Plasma membrane
KEGG Pathway			
FDR	**Associated DEGs**	**Pathway ID**	**Pathways**
0.0087	VWF, SPP1, TNXB, CD36	hsa04512	ECM-receptor interaction
0.0105	JAM2, CLDN5, PECAM1, CDH5	hsa04670	Leukocyte transendothelial migration
0.0158	JAM2, CLDN5, PECAM1, CDH5	hsa04514	Cell adhesion molecules
0.0352	ACADL, FABP4, CD36	hsa03320	PPAR signaling pathway
0.0352	VWF, VEGFD, TEK, SPP1, TNXB	hsa04151	PI3K-Akt signaling pathway

The GO terms ‘vascular process’ and ‘plasma membrane’ are also related to the angiogenesis process in tumor development. Angiogenesis is the process of forming a new vascular network, which is important to transport blood, oxygen, and nutrients to the growing tumor (Liu et al., 2025).

Conclusion

In conclusion, 920 DEGs from the GSE10072 dataset were successfully identified using R, and the 50 top significant genes were found to be related to several pathways, including pathways related to vascular process and plasma membrane.

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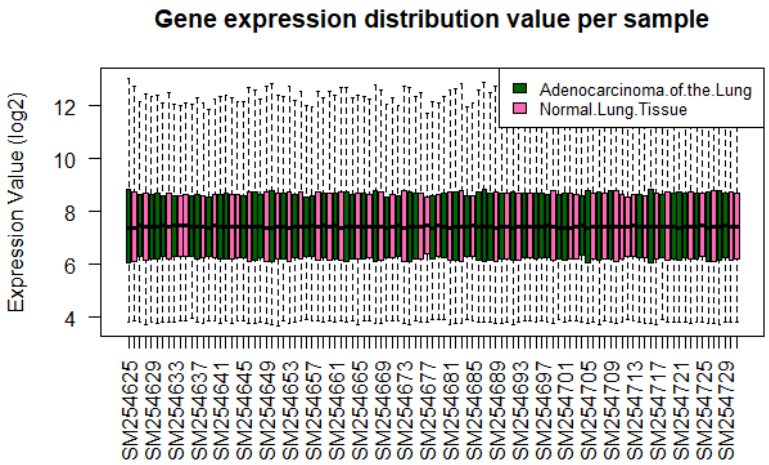
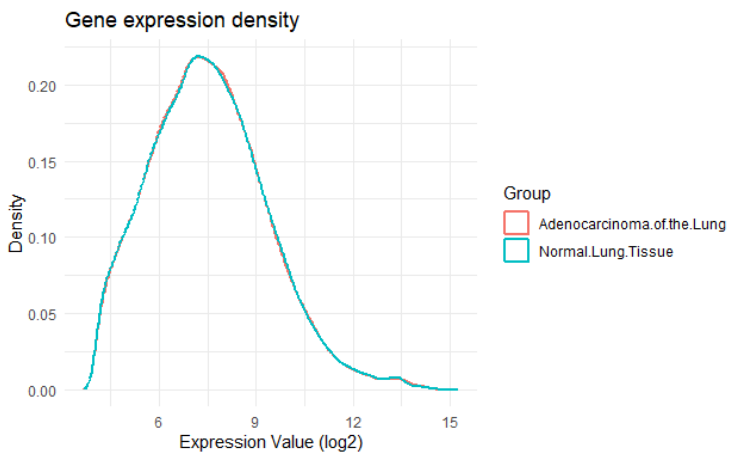
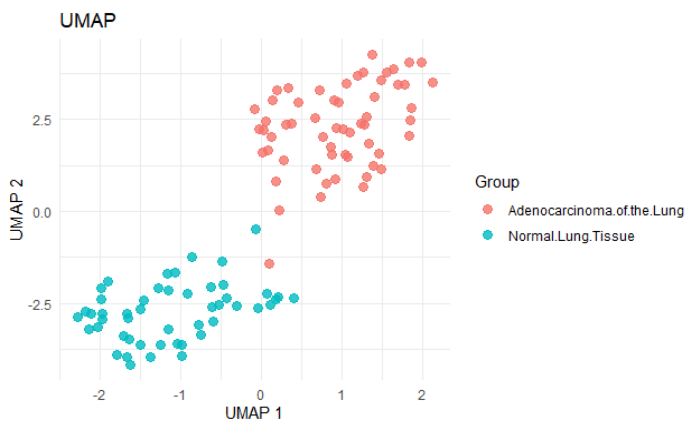
Supplementary



Supplementary 1. Boxplot of GSE10072. The data showed a similar distribution value for each samples.



Supplementary 2. Density plot of GSE10072. The data is right skewed and both groups shared the same density.



Supplementary 3. UMAP plot of GSE10072. The plot showed the clustering between two groups, LUAD (red) and normal lung tissue (blue). There is one LUAD group that is more closely related to the normal lung tissue group.